

Engineering Electromagnetics – Experiment 4

Magnetic Focusing Simulation

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1 Experiment Task Description

This experiment studies the magnetic focusing phenomenon of charged particles in a uniform magnetic field. Sixteen particles start from the origin with the same mass $m = 0.02$ kg and charge $q = 0.016$ C. The electric field is zero,

$$\mathbf{E} = (0, 0, 0),$$

and the magnetic flux density is

$$\mathbf{B} = (0, 0, 8) \text{ T}.$$

The particles share the same initial velocity component along the z -axis,

$$v_z(0) = 10 \text{ m/s},$$

while the transverse components are

$$v_x(0) = 0.1 \sin\left(\frac{k\pi}{8}\right), \quad v_y(0) = 0.1 \cos\left(\frac{k\pi}{8}\right),$$

where $k = 0, 1, 2, \dots, 15$.

The purpose of this experiment is to use MATLAB to simulate the particle trajectories and observe the magnetic focusing effect.

2 Physical Principle

The motion of a charged particle in an electromagnetic field is governed by the Lorentz force:

$$\mathbf{F} = q\mathbf{E} + q(\mathbf{v} \times \mathbf{B}).$$

According to Newton's second law,

$$\mathbf{a} = \frac{\mathbf{F}}{m}.$$

The velocity and position are updated by time discretization:

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \mathbf{a}(t)\Delta t,$$

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t.$$

In this experiment, $\mathbf{E} = 0$ and $\mathbf{B}$ points in the positive z -direction, so each particle moves in a helical path. Since all particles have the same velocity component along the magnetic field, they have nearly the same pitch and can focus again after one cyclotron period. The theoretical angular frequency is

$$\omega = \frac{qB}{m} = \frac{0.016 \times 8}{0.02} = 6.4 \text{ rad/s},$$

so the focusing period is approximately

$$T = \frac{2\pi}{\omega} \approx 0.982 \text{ s}.$$

Therefore, the first focusing position should appear near

$$z_f = v_z T \approx 10 \times 0.982 = 9.82 \text{ m}.$$

3 Simulation Method

In the simulation, the explicit Euler method was used with a time step of $\Delta t = 0.001 \text{ s}$. To observe the motion at different scales, the total simulation time was set to $t = 1, 3, 8, 20$, and 100 s . For each case, three views were plotted:

- a 3D trajectory plot,
- a top view in the x - y plane,
- a front view in the x - z plane.

The program calculates the Lorentz force first, then updates the velocity and position step by step, and finally plots the trajectories of all 16 particles. In this way, the focusing phenomenon can be observed clearly.

4 Experiment Results and Analysis

4.1 Result at $t = 1$ s

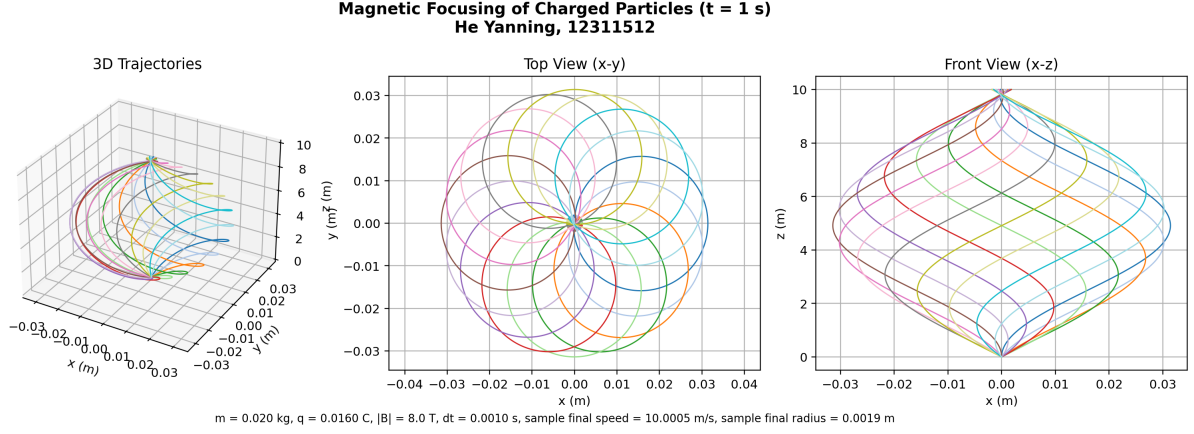


Figure 1: Particle trajectories when the simulation time is 1 s. He Yanning, 12311512

At $t = 1$ s, the particles have completed about one cyclotron period. The front view shows that the particle beam is already close to a focusing point near $z \approx 10$ m, which agrees with the theoretical result. The top view is nearly circular, showing stable helical motion in the transverse plane.

4.2 Result at $t = 3$ s

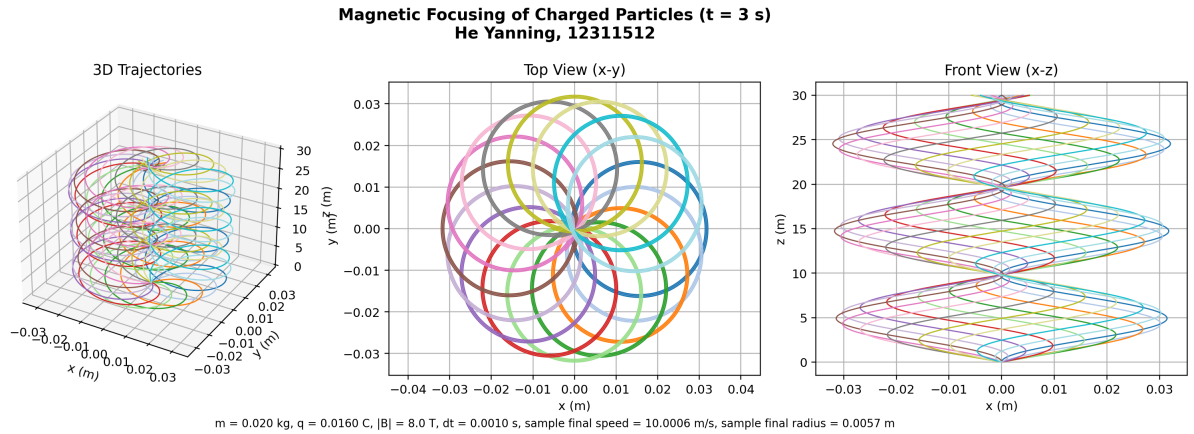


Figure 2: Particle trajectories when the simulation time is 3 s. He Yanning, 12311512

At $t = 3$ s, the repeated helical structure becomes clear. Compared with the result at $t = 1$ s, more than one focusing point can now be observed in the front view. This shows that the focusing process repeats periodically along the z -axis.

4.3 Result at $t = 8$ s

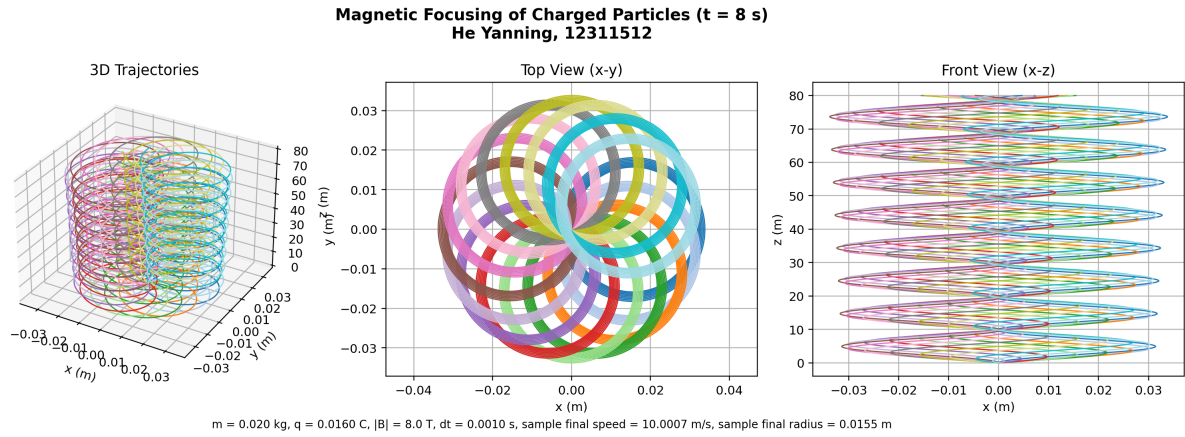


Figure 3: Particle trajectories when the simulation time is 8 s. He Yanning, 12311512

At $t = 8$ s, the periodic focusing behavior becomes more obvious. The front view shows several repeated intersections, while the top view still keeps good rotational symmetry. This indicates that the particles continue to rotate regularly in the transverse plane while moving forward along the magnetic field direction.

4.4 Result at $t = 20$ s

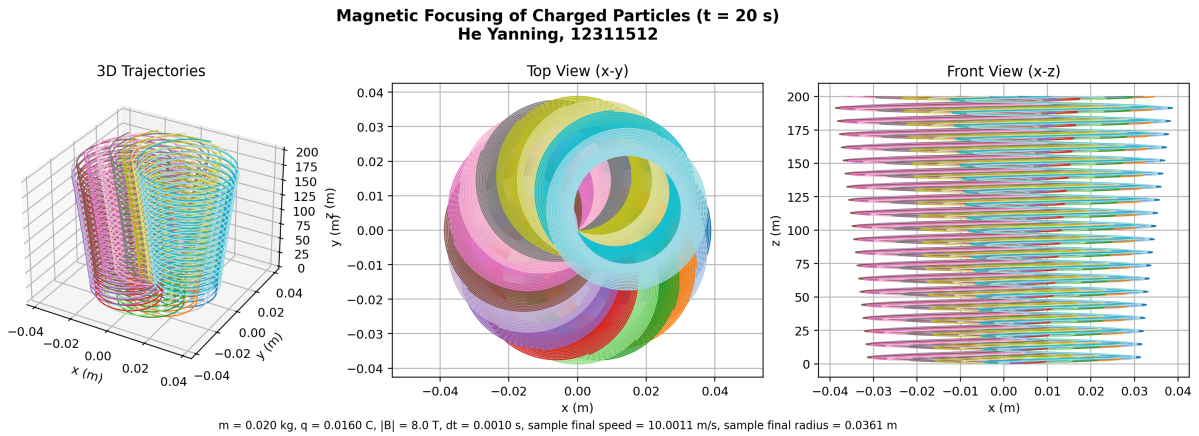


Figure 4: Particle trajectories when the simulation time is 20 s. He Yanning, 12311512

When the simulation time is extended to 20 s, the particle beam shows a regular sequence of focusing points. The spacing between neighboring focusing points is nearly constant, which agrees with the conclusion that the focusing period is determined by the cyclotron motion.

4.5 Result at $t = 100$ s

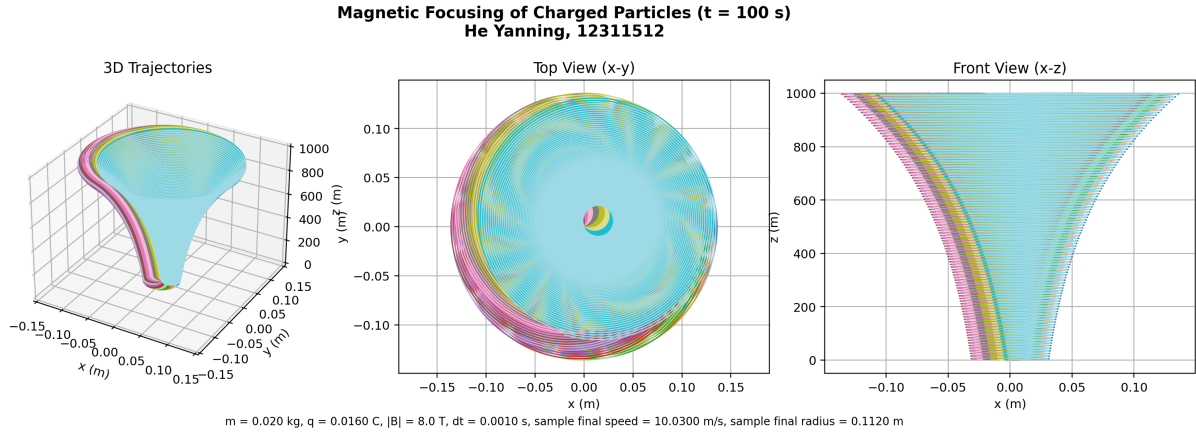


Figure 5: Particle trajectories when the simulation time is 100 s. He Yanning, 12311512

At $t = 100$ s, the beam forms a long helical bundle along the magnetic-field direction. Even at this large time scale, the repeated focusing points are still visible. This shows that particles with the same axial velocity can keep synchronous focusing for a long time in a uniform magnetic field.

4.6 Summary of Observations

Item	Observation
Motion type	Helical motion around the z -axis
Focusing mechanism	Same v_z and same cyclotron period
First focusing position	Near $z \approx 9.82$ m
Transverse pattern	Approximately circular and symmetric
Long-term behavior	Periodic reconvergence along the field direction

Table 1: Main observations from the simulation.

5 MATLAB Source Code

The MATLAB source code used in this experiment is listed below. Detailed Chinese comments are included to explain the physical parameters, the numerical calculation process, and the plotting procedure.

```

1 clc;
2 clear;
3 close all;

```

```

4
5 % 实验四：带电粒子在磁场中的聚焦运动
6 % 姓名：何衍宁
7 % 学号：12311512
8
9 % =====
10 % 一、物理参数设置
11 % =====
12 m = 0.02; % 粒子质量，单位 kg
13 q = 1.6e-2; % 粒子电荷量，单位 C
14 E = [0, 0, 0]; % 电场强度矢量，题设中取 0
15 B = [0, 0, 8]; % 磁通密度矢量，沿 z 轴方向
16 dt = 0.001; % 时间步长，单位 s
17 t_ends = [1, 3, 8, 20, 100]; % 选取不同观察时长，便于从不同尺度分析轨迹
18 num_particles = 16; % 粒子束中粒子个数
19 output_dir = fullfile(pwd, 'figures'); % 图像输出目录
20
21 % 若输出目录不存在，则自动创建
22 if ~exist(output_dir, 'dir')
23     mkdir(output_dir);
24 end
25
26 % =====
27 % 二、对不同时间尺度分别作图
28 % =====
29 for case_idx = 1:numel(t_ends)
30     t_end = t_ends(case_idx); % 当前仿真的终止时间
31     t = 0:dt:t_end; % 构造时间数组
32
33     % 创建一张图，内部用三联图显示三维图、俯视图和侧视图
34     fig = figure('Color', 'w', 'Position', [100, 100, 1400, 420]);
35     tl = tiledlayout(fig, 1, 3, 'TileSpacing', 'compact', 'Padding',
        ↳ 'compact');
36     title(tl, sprintf('Magnetic Focusing of Charged Particles (t = %g
        ↳ s)\nHe Yanning, 12311512', t_end), ...
37         'FontSize', 14, 'FontWeight', 'bold');
38
39     % -----
40     % 1. 三维轨迹图
41     % -----
42     ax1 = nexttile;

```

```

43     hold(ax1, 'on');
44     grid(ax1, 'on');
45     view(ax1, 3);
46     xlabel(ax1, 'x (m)');
47     ylabel(ax1, 'y (m)');
48     zlabel(ax1, 'z (m)');
49     title(ax1, '3D Trajectories');
50
51     % -----
52     % 2. 俯视图 (x-y 平面)
53     % -----
54     ax2 = nexttile;
55     hold(ax2, 'on');
56     grid(ax2, 'on');
57     axis(ax2, 'equal');
58     xlabel(ax2, 'x (m)');
59     ylabel(ax2, 'y (m)');
60     title(ax2, 'Top View (x-y)');
61
62     % -----
63     % 3. 侧视图 (x-z 平面)
64     % -----
65     ax3 = nexttile;
66     hold(ax3, 'on');
67     grid(ax3, 'on');
68     xlabel(ax3, 'x (m)');
69     ylabel(ax3, 'z (m)');
70     title(ax3, 'Front View (x-z)');
71
72     % 配色方案: 为 16 个粒子分配不同颜色, 便于区分轨迹
73     cmap = lines(num_particles);
74
75     % =====
76     % 三、逐粒子计算运动轨迹
77     % =====
78     for k = 0:(num_particles - 1)
79         % 设置第 k 个粒子的初速度
80         % z 方向分量相同, x-y 平面分量按题目给定的正弦余弦方式分布
81         v0 = [0.1 * sin(k * pi / 8), 0.1 * cos(k * pi / 8), 10];
82
83         % 调用子函数, 用欧拉法计算粒子轨迹

```

```

84     [r, v] = simulate_particle(m, q, E, B, dt, t, v0);
85
86     % 在三维图、俯视图和侧视图中分别绘制轨迹
87     plot3(ax1, r(:, 1), r(:, 2), r(:, 3), 'Color', cmap(k + 1, :),
88           ↪ 'LineWidth', 1.1);
89     plot(ax2, r(:, 1), r(:, 2), 'Color', cmap(k + 1, :), 'LineWidth',
90           ↪ 1.1);
91     plot(ax3, r(:, 1), r(:, 3), 'Color', cmap(k + 1, :), 'LineWidth',
92           ↪ 1.1);
93
94     % 记录第一条轨迹的末速度和末半径，便于在图下方添加说明
95     if k == 0
96         final_speed = norm(v(end, :));
97         final_radius = hypot(r(end, 1), r(end, 2));
98     end
99 end
100
101 % 图下方附加简短说明，展示参数与计算结果
102 annotation(fig, 'textbox', [0.34, 0.01, 0.32, 0.07], ...
103           'String', sprintf('m = %.3f kg, q = %.4f C, |B| = %.1f T, dt = %.4f
104           ↪ s, sample final speed = %.4f m/s, sample final radius = %.4f
105           ↪ m', ...
106           m, q, norm(B), dt, final_speed, final_radius), ...
107           'HorizontalAlignment', 'center', 'EdgeColor', 'none', 'FontSize',
108           ↪ 10);
109
110 % 将当前图保存到 figures 文件夹中
111 exportgraphics(fig, fullfile(output_dir,
112           ↪ sprintf('magnetic_focusing_t%.g.png', t_end)), 'Resolution', 220);
113 close(fig);
114 end
115
116 fprintf('Finished generating figures in %s\n', output_dir);
117
118 % =====
119 % 子函数：用显式欧拉法求解单个粒子的运动过程
120 % 输入：
121 %   m, q : 粒子质量和电荷量
122 %   E, B : 电场和磁场矢量
123 %   dt   : 时间步长
124 %   t    : 时间数组

```



```

118 % v0 : 初速度
119 % 输出:
120 % r : 每个时刻的位置矢量
121 % v : 每个时刻的速度矢量
122 % =====
123 function [r, v] = simulate_particle(m, q, E, B, dt, t, v0)
124 num_steps = numel(t);
125 r = zeros(num_steps, 3); % 位置数组, 三列分别对应 x、y、z
126 v = zeros(num_steps, 3); % 速度数组, 三列分别对应 x、y、z
127 v(1, :) = v0; % 设置初始速度
128
129 for i = 1:(num_steps - 1)
130 % 计算洛伦兹力:  $F = qE + q(v \times B)$ 
131 force = q * E + q * cross(v(i, :), B);
132
133 % 根据牛顿第二定律求加速度:  $a = F / m$ 
134 accel = force / m;
135
136 % 显式欧拉法更新下一时刻速度
137 v(i + 1, :) = v(i, :) + accel * dt;
138
139 % 显式欧拉法更新下一时刻位置
140 r(i + 1, :) = r(i, :) + v(i, :) * dt;
141 end
142 end

```

6 Experiment Experience

At the beginning, I only understood magnetic focusing from the formula. After plotting the trajectories, I could see much more clearly how the particles spiral forward and gather again after one period. This made the physical meaning of the result easier to understand.

During the simulation, I also found that the choice of time step and total simulation time is very important. If the time step is too large, the trajectory is not accurate enough. If the observation time is too short, only the local motion can be seen; if it is longer, the repeated focusing process becomes much easier to observe.

Through this experiment, I gained a better understanding of the Lorentz force, helical motion in a uniform magnetic field, and the reason why magnetic focusing occurs. I also became more familiar with using MATLAB to solve electromagnetic problems with numerical methods.